

Review article

Fluorescence-guided glioma surgery improves intraoperative visualization and resection without increasing neurological morbidity: A systematic review and meta-analysis

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ABSTRACT

Background: Fluorescence-guided surgery (FGS) has been increasingly adopted to improve intraoperative tumor visualization and the extent of resection in high-grade gliomas. However, concerns remain about whether enhanced visualization and more aggressive resection could lead to increased postoperative neurological deficits. We conducted a systematic review and meta-analysis to assess the neurological safety of fluorescence-guided surgery compared with conventional white-light (WL) surgery.

Methods: A systematic search of PubMed, Embase, and Scopus was performed from database inception to March 2026, following PRISMA guidelines. Studies comparing FGS (such as 5-aminolevulinic acid or fluorescein) with WL surgery and reporting postoperative neurological outcomes were included.

Results: A total of 27 studies involving 3241 patients (1974 undergoing FGS and 1267 undergoing WL) were included in the systematic review. Of these, 13 comparative studies were suitable for a quantitative analysis of postoperative neurological deficits. The combined analysis showed no significant difference in neurological deficit risk between FGS and WL surgery (RR=1.02, 95% CI 0.75–1.40; $I^2 = 39.3\%$). Sensitivity analysis, excluding studies with zero events, produced similar results (RR=1.03, 95% CI 0.71–1.51; $I^2 = 54.4\%$). Subgroup analysis by study design showed no significant differences between randomized controlled trials and observational studies (p for subgroup difference >0.05). The overall risk of bias was moderate, mainly due to confounding in observational studies, and the evidence was rated as low certainty.

Conclusion: In our analysis, FGS does not lead to more postoperative neurological deficits when compared to traditional white-light surgery. These results support its use as a safe addition to improve tumor visualization and enable maximal removal without compromising neurological function. Future research involving patient-reported outcomes is necessary to better understand the patient-centered benefits of fluorescence-guided techniques.

1. Introduction

Gliomas represent the most common primary malignant tumors of the central nervous system and are broadly classified into low-grade (WHO grades I–II) and high-grade (WHO grades III–IV) lesions based on histopathological and molecular features [1]. High-grade gliomas (HGGs), particularly glioblastoma multiforme (GBM), are characterized by aggressive biological behavior, diffuse infiltration, and poor clinical outcomes despite advances in multimodal therapy [2–4]. Although GBM

rarely metastasizes beyond the central nervous system, its highly infiltrative nature within the brain parenchyma poses a significant challenge to achieve durable local control [1–3].

Maximal safe surgical resection remains a cornerstone of HGG management, typically followed by adjuvant chemotherapy and radiotherapy [1,2,5,6]. However, the indistinct boundaries between tumor and normal brain tissue often limit the extent of resection achievable with conventional techniques. In this context, the balance between achieving maximal tumor removal and preserving neurological function

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is a central challenge in neurosurgical oncology [3,7,8].

Conventional white-light (WL) microscopy has long served as the standard intraoperative visualization modality. While widely available and cost-effective, this approach relies heavily on subjective visual cues and surgeon experience, which may contribute to incomplete tumor resection and residual disease [3,7–9]. To overcome these limitations, fluorescence-guided surgery (FGS), particularly using 5-aminolevulinic acid (5-ALA), has been increasingly adopted. By enhancing intraoperative tumor visualization, FGS has been associated with improved extent of resection and progression-free survival in randomized and observational studies [8]. However, fluorescence-guided surgery introduces an important clinical trade-off. While improved visualization may facilitate more extensive tumor resection, particularly near eloquent cortex, it may also increase the risk of postoperative neurological deficits, underscoring the need to balance oncologic benefit with functional preservation [3,7–9]. While prior studies have largely focused on oncologic endpoints such as extent of resection and survival, the impact of fluorescence-guided techniques on neurological morbidity and functional outcomes remains incompletely defined and inconsistently reported [8–10].

Given this critical gap, a comprehensive evaluation of neurological safety is warranted. Therefore, we conducted a systematic review and meta-analysis comparing FGS with conventional WL microscopy for high-grade glioma resection, with a specific focus on postoperative neurological deficits and functional outcomes.

2. Methods

This systematic review and meta-analysis were conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines. The study protocol was prospectively developed prior to data extraction and registered with PROSPERO (Registration Number: 1360961). Data supporting the findings of this study are available from the corresponding author upon reasonable request.

2.1. Search strategy and eligibility criteria

A comprehensive systematic search of the literature was performed to identify studies comparing FGS with conventional WL microsurgery for intracranial tumor resection. The following electronic databases were searched from inception through March 2026: PubMed (MEDLINE), Embase, and Scopus.

The search strategy incorporated both controlled vocabulary and free-text terms related to fluorescence-guided surgery and neurosurgical oncology. Key search terms included combinations of: “fluorescence-guided surgery,” “5-ALA,” “aminolevulinic acid,” “fluorescein,” “white light,” “glioma,” “brain tumor,” and “neurosurgery” and functional outcomes (including “neurological deficit,” “neurological deterioration,” “functional outcome,” and “Karnofsky Performance Status”). Boolean operators were used to optimize sensitivity and specificity. The complete search strategy for each database is provided in [Supplementary Table S1](#).

Studies were eligible if they: (1) included patients undergoing intracranial tumor resection;

(2) compared FGS (e.g., 5-ALA or fluorescein sodium) with conventional WL surgery; (3) reported postoperative neurological outcomes, specifically new or worsened neurological deficits; and (4) were prospective or retrospective comparative studies, including clinical trials and observational cohorts. We also included studies not comparing FGS with WL that reported new neurological deficits for systematic review purpose. Studies were excluded if they were case reports, case series with fewer than ten patients, review articles, editorials, conference abstracts without full data, preclinical or animal studies, or lacked extractable neurological outcome data. In cases of overlapping cohorts, the most comprehensive or recent study was included to avoid

duplication. Detailed inclusion and exclusion criteria are provided in [Supplementary Table 2](#).

2.2. Study selection and data collection

Study screening and selection were conducted in accordance with the PRISMA 2020 guidelines. All records identified through database searches were imported into EndNote X9 for initial management, where duplicate entries were removed using a combination of automated and manual methods. The deduplicated dataset was subsequently transferred to Mendeley and exported in RIS format for systematic screening using the Rayyan platform. Two independent reviewers (S.S. and M.A.) screened titles and abstracts to identify potentially eligible studies. Articles meeting initial eligibility criteria underwent full-text review for final inclusion. Discrepancies at any stage of the screening process were resolved through consensus, with senior investigators serving as adjudicators when necessary.

To ensure completeness of the search, the reference lists of all included studies and relevant review articles were manually screened to identify additional studies not captured through the electronic search. The study selection process is summarized in the PRISMA flow diagram ([Fig. 1](#)). Data extraction was carried out independently by two investigators (S.S. and M.A.) using a standardized data extraction form developed before the review process. Extracted data included study characteristics (author, year, and design), patient sample sizes, type of fluorescence agent used, and the number of postoperative neurological deficits and total patients in each group. The primary outcome was postoperative neurological deficit, defined as any new or worsened neurological impairment following surgery. When multiple definitions were reported, the most clinically relevant postoperative outcome was extracted. Secondary outcomes included functional status measures (i.e., Karnofsky Performance Status) and transient versus permanent neurological deficits when reported.

2.3. Outcome definition

The primary outcome of interest was the occurrence of new or worsened postoperative neurological deficits, as defined by each individual study. When studies reported multiple time points, the earliest postoperative assessment was preferentially used to ensure consistency across studies. Functional outcomes, including modified Rankin Scale, Karnofsky Performance Status (KPS), and quality-of-life measures, were not included in the quantitative synthesis due to inconsistent reporting, heterogeneity in outcome definitions, and lack of standardized assessment across studies.

2.4. Risk of bias and certainty assessment

The methodological quality of included studies was assessed using established tools in accordance with the Cochrane Handbook. The Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) tool was applied for observational studies, and appropriate tools were used for randomized trials when applicable.

The overall certainty of evidence was evaluated using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework, considering risk of bias, inconsistency, indirectness, imprecision, and publication bias.

2.5. Statistical analysis

Meta-analysis was performed using a random-effects model to account for expected clinical and methodological heterogeneity across studies. Pooled estimates of neurological deficit rates were calculated using Risk Ratio (RR) with corresponding 95% confidence intervals (CIs). Heterogeneity was assessed using Cochran’s Q test and the I² statistic, with values greater than 75% indicating substantial

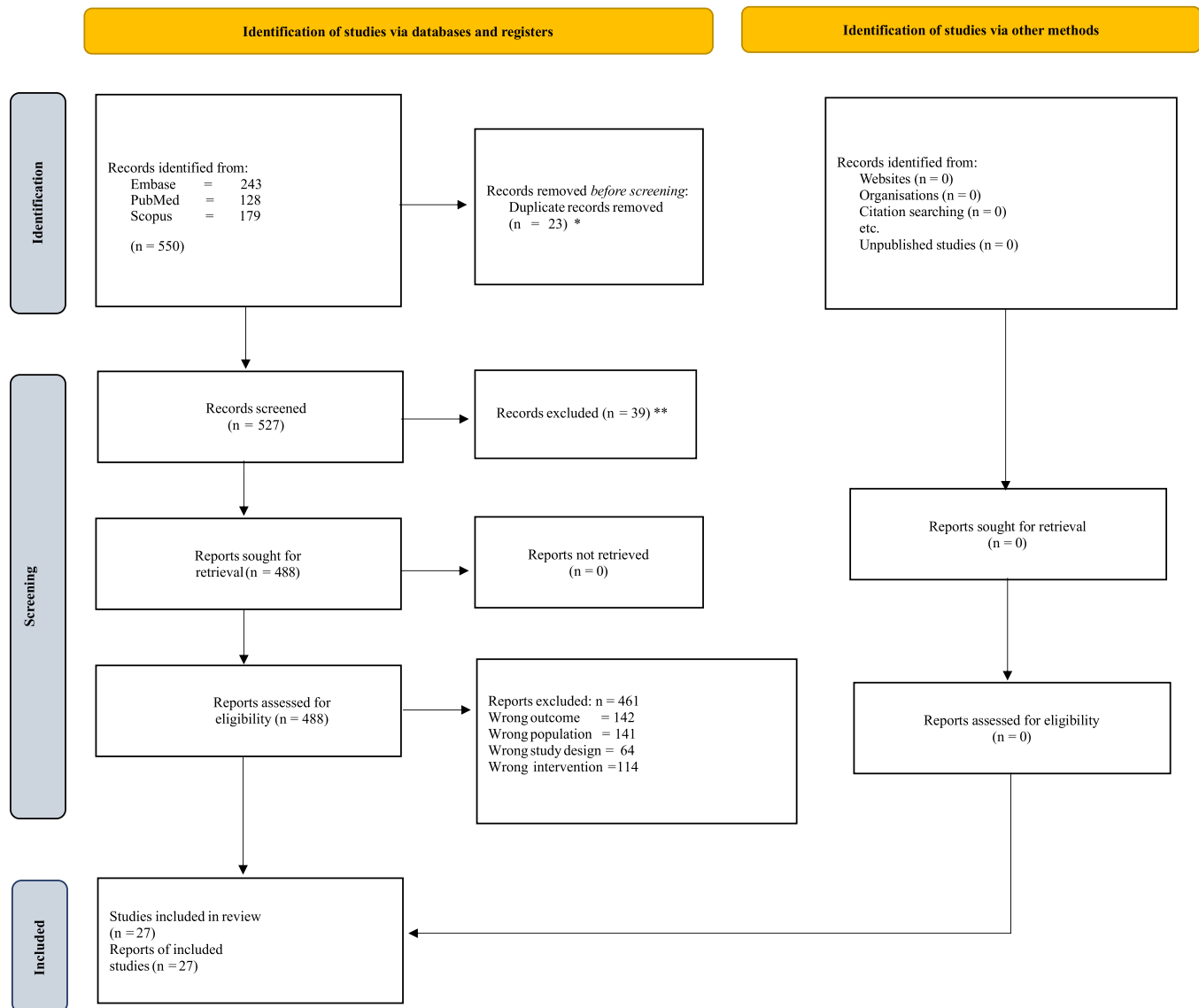


Fig. 1. PRISMA flowchart of study selection. * With automation tool; ** Without automation tool.

heterogeneity. Sensitivity analyses were performed by sequential exclusion of individual studies that reported zero events in FGS and WL resection groups. All statistical analyses were performed using R statistical software (version 4.3.2), and a two-sided p -value < 0.05 was considered statistically significant.

3. Results

Following systematic screening, a total of 488 full-text articles were assessed for eligibility (Fig. 1). After applying predefined inclusion and exclusion criteria, 27 studies [8,11–36] comprising 3241 patients (1974 (60.9%) in FGS and 1267 (39.1%) in WL) were included in the final systematic review (Table 1). Among these, studies that reported both FGS and WL cohorts reporting neurological deficits (Thirteen studies, 48.1%) were included in the quantitative synthesis for meta-analysis [8, 11,13–15,21,22,27–29,32,34,35]. Across the included literature, study designs were heterogeneous, encompassing randomized controlled trials, prospective cohort studies, and retrospective observational studies, with the majority consisting of single-center retrospective cohorts. The included studies were conducted across diverse geographic regions, reflecting global clinical practice in glioma surgery, including Europe (notably Germany, France, Spain, and Italy), Asia (China, Hong Kong,

Malaysia, and Kazakhstan), North America (United States), and the Middle East (Israel). Enrollment periods ranged from the mid-1990s to 2023, capturing both early and contemporary experiences with fluorescence-guided techniques.

FGS was predominantly performed using 5-ALA, which was the most frequently evaluated agent across studies. A smaller subset of studies incorporated alternative fluorescence techniques, including fluorescein-based approaches, highlighting evolving intraoperative visualization strategies. Across comparative studies included in the meta-analysis. Baseline patient characteristics were generally comparable across studies, with mean ages typically ranging from the mid-40s to early 60 s, consistent with the epidemiology of high-grade gliomas [37]. Functional and neurological outcomes were most frequently assessed using standardized clinical scales, including the KPS and National Institutes of Health Stroke Scale (NIHSS), while several studies also incorporated radiographic endpoints such as extent of resection (EOR) assessed by volumetric magnetic resonance imaging.

Definitions of neurological outcomes varied across studies, although most reported new or worsened postoperative neurological deficits or deterioration in functional status. Some studies additionally differentiated between transient and permanent deficits or incorporated threshold-based functional decline (i.e., postoperative KPS reduction).

Table 1
Characteristics of Included Studies Evaluating Fluorescence-Guided Versus White-Light Surgery in High-Grade Gliomas.

Study	Country	Study design	Enrollment period	Total number of patients overall	No of Patients in fluorescent group	No of Females in fluorescent group	No of Patients in white light group	No of Females in white light	Age of study population	Scale used	No of events (neurological deficits)- fluorescent	No of patients fluorescent	No of events (neurological deficits)- white light	No of patients white light
Stummer et al., 2006	Germany	Controlled trial	Oct 11, 1999-July 19, 2004	322	161	NR	161	NR	Fluorescent-60, white light-59	KPS, NIH Stroke scale, MRI volumetrics	10	139	3	131
Mirza et al., 2022	UK	Single-center retrospective cohort study	5-ALA group: Jan 2013-Jan 2019; White-light group: Jan 2009-2013	69	39	14	30	18	5-ALA: 46.77 $\pm$ 1.74; White-light: 41.77 $\pm$ 2.42	Performance Status (PS), 6-month PS (PS6m)	3	39	1	30
Chan et al., 2018	Hong Kong	Retrospective cohort study	2011-2016	165	15	12	N/A	N/A	48.31 years (SD 14.70)	No functional or symptom scale used; only extent of resection (EOR) categories (total, >95%, >90%).	3	15	N/A	N/A
Ling et al., 2024	China	Prospective randomized controlled trial + retrospective white-light cohort	June 2020 – June 2022 (FL groups); white-light cases from before June 2020	90	60	26	30	18	1 mg/kg: 44.76 $\pm$ 10.68; 5 mg/kg: 50.06 $\pm$ 12.11; WL: 49.20 $\pm$ 13.09	KPS, 6-month PFS, EOR volumetrics; GFAP & Ki-67 IHC	7	60	5	30
Picart et al., 2017	France	Single-center retrospective consecutive cohort study	November 2012 – November 2015	51	24	NR	27	NR	5-ALA: 56.9 $\pm$ 11.9, White-light: 65.8 $\pm$ 11.8	KPS	11	24	7	27
Chen et al., 2012	China	Prospective study	June 2010 – June 2011	22	10	6	12	5	5-ALA: 50.7 $\pm$ 10.4; White-light: 51.8 $\pm$ 6.8	KPS	3	10	5	12
Pichlmeier et al., 2008	Germany	Secondary analysis of RCT (post-hoc RPA survival analysis)	1999-2004 (ALA trial period)	243	122	NR	121	NR	NR	KPS and NIH-SS	N/A	61	N/A	51
Teixidor et al., 2016	Spain	Prospective multicenter cohort	May 2010 – September 2014	77	77	34	N/A	N/A	Mean 57.7 years (SD 13.8)	KPS	14	77	N/A	N/A
Stummer et al., 2000	Germany	Prospective cohort study	December 1995 – December 1998	52	52	16	N/A	N/A	Mean 53.8 $\pm$ 10.1 years (range 31-69)	KPS	1	52	N/A	N/A

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Table 1 (continued)

Study	Country	Study design	Enrollment period	Total number of patients overall	No of Patients in fluorescent group	No of Females in fluorescent group	No of Patients in white light group	No of Females in white light group	Age of study population	Scale used	No of events (neurological deficits)- fluorescent	No of patients fluorescent	No of events (neurological deficits)- white light	No of patients white light
Valle et al., 2011	Spain	Prospective single-center cohort study	October 2007 – June 2009	36	36	NR	N/A	N/A	58 years	KPS	2	36	N/A	N/A
Tejada-Solis et al., 2012	Spain	Prospective observational cohort	Aug 2007 – Jan 2012	28	28	NR	N/A	N/A	58.5 years	KPS	2	28	N/A	N/A
Stummer et al., 2011	Germany	Randomized multicenter Phase III controlled trial	NR	349	176	74	173	62	~58.6 years	NIH-SS, KPS	22	176	20	173
Laviv et al., 2025	Israel	Retrospective cohort	2011–2021	94	48	NR	46	NR	60.45 years	KPS (primary), EOR, PFS, OS	0	48	0	46
Sloty et al., 2013	Germany	Retrospective single-center cohort	2002–2009	253	68	NR	185	NR	58.7 years	KPS, MRI-based extent of resection, MGMT status	19	61	13	51
Zeppa et al., 2022	Italy	Single-center retrospective analysis	Jan 2018 – Jan 2021	99	99	50	N/A	N/A	5-ALA: 63.9; SF: 60.8; Both: 61.7	KPS	0	99	N/A	N/A
He et al., 2020	China	Prospective clinical trial	April 2016 – August 2018	29	29	NR	N/A	N/A	Median 54 (range 17–70)	KPS	0	29	N/A	N/A
Hohne et al., 2020	Germany, Italy, Turkey	Multicenter retrospective study	May 2012 – Feb 2020	18	18	13	N/A	N/A	Mean 22.9 years (range 3–78)	KPS	5	18	N/A	N/A
Ryskeldiev et al., 2025	Kazakhstan	Retrospective cohort	2018–2023	141	71	27	70	20	48.8 years (overall mean)	KPS	Postoperative KPS ≤ 70: 39 events	71	Postoperative KPS ≤ 70: 49 events	70
Picart et al., 2023	France	Phase III RCT, prospective, single-blinded, multicenter	2013–2016	171	88	23	83	26	60.7 vs 58.2 yrs	KPS	9	74	9	73
Xi et al., 2023	China	Single-center retrospective cohort	July 2017 – June 2022	112	61	27	51	18	Fluor: 53.20 ± 13.38 yrs; WL: 51.63 ± 16.89 yrs	RANO criteria for EOR; Clavien-Dindo for complications	0	61	0	51
Muscas et al., 2022	Italy	Retrospective analysis of a prospectively collected database	June 2016 – June 2019	65	65	35	N/A	N/A	56.9 ± 10.0 years	MRC Motor Scale for neurological outcome; CRET for radiological outcome	1	65	Not applicable	Not applicable
Suhupper et al., 2021	United States	Prospective multicenter single-arm study	June 9, 2016 – December 24, 2018	69	69	27	N/A	N/A	60.5 ± 12.4 years	KPS (neurological function), CTCAE (adverse events), CRET/EOR (radiological)	3	69	N/A	N/A

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Table 1 (continued)

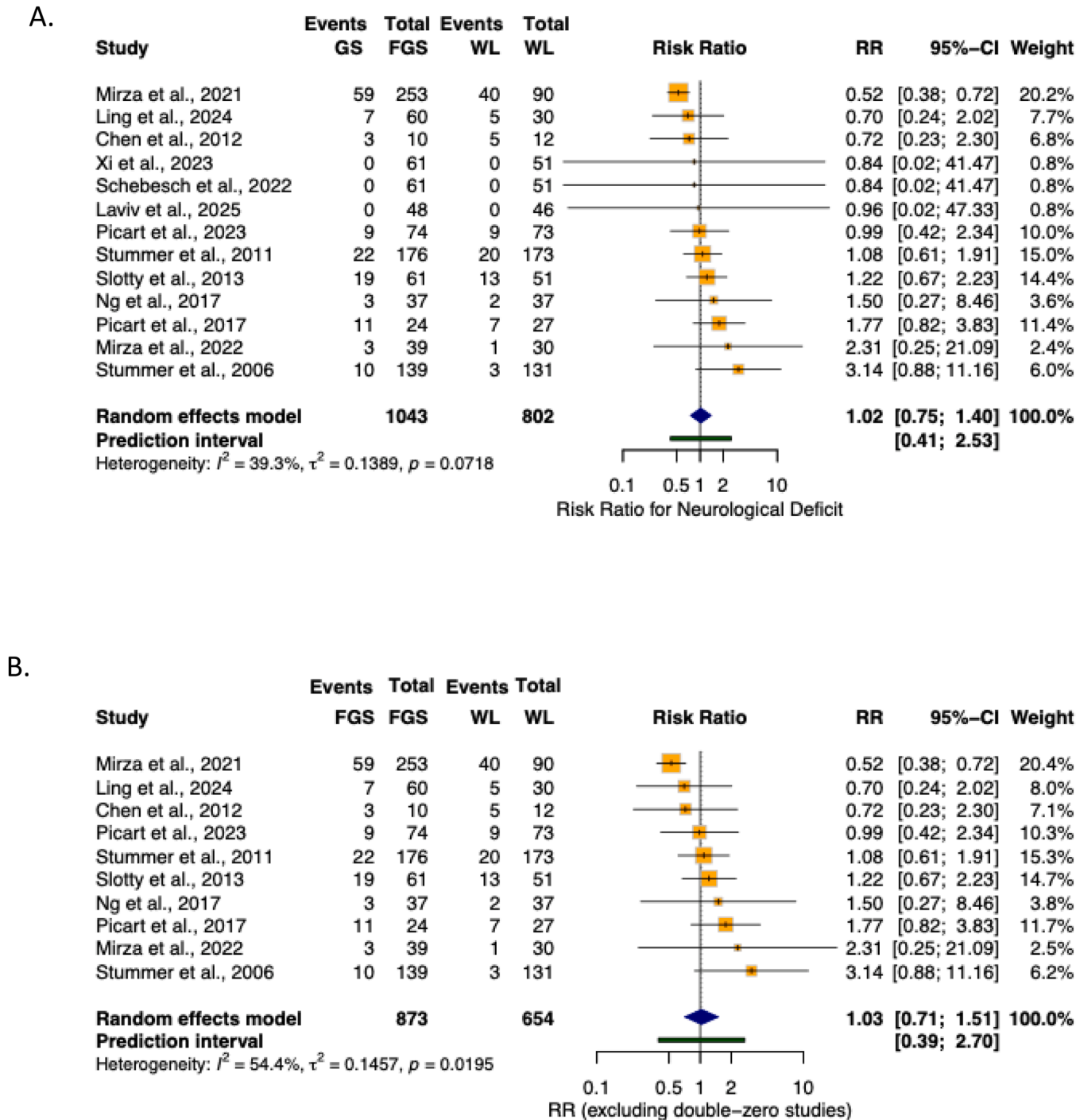
Study	Country	Study design	Enrollment period	Total number of patients overall	No of Patients in fluorescent group	No of Females in fluorescent group	No of Patients in white light group	No of Females in white light	Age of study population	Scale used	No of events (neurological deficits)- fluorescent	No of patients fluorescent	No of events (neurological deficits)- white light	No of patients white light
Schebesch et al., 2022	Germany	Retrospective analysis of a prospective registry	2013–2022 (registry began 2013; all included patients newly diagnosed HGG)	347	196	77	151	64	Median age 62.4 years	KPS (neurological performance), radiological volumetrics (EOR, residual volume), PFS/ OS via Kaplan–Meier	0	61	0	51
Kutlay et al., 2021	Turkey	Retrospective review of prospectively collected cases	2016–2020	20	20	8	N/A	N/A	Mean 51.2 years (range 18–78)	Karnofsky Performance Status (KPS); Extent of Resection (GTR/NTR/ STR)	0	20	N/A	N/A
Mirza et al., 2021	United Kingdom	Single-center retrospective cohort study	2013–2019 (fluorescent cohort); 2009–2013 (historical white-light cohort)	343	253	79	90	33	5-ALA: 56.69 ± 0.75 yrs; White-light: 54.43 ± 1.73 yrs	Performance Status (PS), new focal neurological deficit, radiological EoR (GTR/ STR), overall survival	59	253	40	90
Ng et al., 2017	Malaysia	Retrospective cohort study	January 2008 – December 2014	74	37	14	37	10	FG: 49.6 ± 11.6 yrs; WL: 49.2 ± 12.8 yrs	Karnofsky Performance Scale (KPS), survival time (months)	3	37	2	37
Aldave et al., 2013	Spain	Retrospective cohort study	August 2007 – December 2011	52	52	23	Not applicable	Not applicable	Mean age: 56.2 (no residual) vs 61.2 (residual)	Neurological deficit assessment; Karnofsky Performance Scale (KPS); MRI-based CRET; survival (OS)	6	52	Not applicable	Not applicable

Table summarizing the characteristics of included studies comparing fluorescence-guided surgery (FGS) and conventional white-light (WL) microscopy in patients with high-grade gliomas. Data include study design, geographic location, enrollment period, patient demographics, surgical modality, outcome measures, and reported neurological deficits. FGS, fluorescence-guided surgery; WL, white-light; 5-ALA, 5-aminolevulinic acid; KPS, Karnofsky Performance Status; NIHSS, National Institutes of Health Stroke Scale; EOR, extent of resection; MRI, magnetic resonance imaging; PFS, progression-free survival; OS, overall survival; RCT, randomized controlled trial; NR, not reported; IHC, immunohistochemistry; RANO, Response Assessment in Neuro-Oncology; CTCAE, Common Terminology Criteria for Adverse Events; MRC, Medical Research Council; CRET, contrast-enhanced resection tumor; GTR, gross total resection; STR, subtotal resection.

Despite variability in reporting, neurological morbidity was consistently evaluated across both fluorescence-guided and white-light cohorts, enabling comparative synthesis. A comprehensive summary of study characteristics, including study design, patient demographics, surgical techniques, and outcome definitions, is provided in Table 1.

3.1. New neurological deficits following surgery

A total of 13 comparative studies were included in the primary meta-analysis of postoperative neurological deficits, comprising 1845 patients overall, including 1043 patients in the FGS group and 802 patients in the WL group. Pooled random-effects analysis demonstrated no significant difference in the risk of postoperative neurological deficits between FGS and WL surgery (RR 1.02, 95% CI [0.75–1.40]; $I^2 = 39.3\%$) (Fig. 2A)



Sensitivity analysis excluding studies with zero events in both treatment arms yielded consistent findings, again showing no significant difference in neurological deficit risk between groups (RR 1.03, 95% CI [0.71–1.51]; $I^2 = 54.4\%$). Overall, these findings indicate that fluorescence-guided resection was associated with a comparable postoperative neurological safety profile to conventional white-light surgery (Fig. 2B).

3.2. Subgroups analysis based on study design

Subgroup analysis based on study design demonstrated no significant differences in neurological outcomes between FGS and WL surgery across both randomized controlled trials (RCTs) and observational studies (OS). Among RCTs, pooled analysis showed no significant association between FGS and postoperative neurological deficits (RR 1.11, 95% CI [0.55–2.25]; $I^2 = 11.5\%$). Similarly, in observational studies, no significant difference was observed (RR 0.98, 95% CI [0.65–1.48]; $I^2 = 41.4\%$). There was no evidence of a subgroup effect based on study design (p for subgroup differences = 0.6656) (Fig. 3A).

Sensitivity analysis excluding studies with zero events yielded consistent findings. In this restricted cohort, RCTs continued to show no significant difference (RR 1.11, 95% CI [0.55–2.25]; $I^2 = 11.5\%$), while observational studies also demonstrated no significant association (RR 0.99, 95% CI [0.55–1.79], $I^2 = 63.3\%$). The test for subgroup differences remained non-significant ($p = 0.7299$), indicating no differential effect between study designs (Fig. 3B).

3.3. Risk of bias and certainty

Risk of bias was assessed using the Cochrane Risk of Bias 2 (RoB 2) tool for randomized controlled trials and the ROBINS-I tool for observational studies. Randomized trials were generally judged to have low risk to some concerns, primarily related to deviations from intended interventions. Observational studies were assessed as having moderate risk of bias, largely due to potential confounding by indication, as fluorescence-guided surgery may have been preferentially used in more complex or eloquent tumors. Overall, while most domains demonstrated low risk, the inclusion of non-randomized studies limits the overall methodological certainty (Supplementary Figure 1 and 2)

The certainty of evidence for postoperative neurological outcomes was assessed as low according to the GRADE framework. This was primarily due to risk of bias from observational study designs and imprecision, as confidence intervals were wide and crossed the line of no effect. Although findings were consistent across primary and sensitivity analyses, these methodological limitations reduce confidence in the precision of the estimated effect (Supplementary Table 3).

4. Discussion

In this systematic review and meta-analysis, we found that FGS was not associated with an increased risk of postoperative neurological deficits compared with conventional WL surgery. The pooled analysis demonstrated comparable neurological outcomes between groups, with consistent findings across sensitivity analyses excluding studies with zero events. Subgroup analyses further confirmed that this effect was stable across both randomized controlled trials and observational studies, with no evidence of interaction by study design. Collectively, these results indicate that fluorescence-guided resection does not compromise neurological safety.

Maximal safe resection remains a central principle in the surgical management of high-grade gliomas, with extent of resection strongly associated with improved oncologic outcomes [7,38–40]. However, increasing the extent of resection raises a fundamental concern regarding potential neurological morbidity, particularly in eloquent brain regions [41–43]. Fluorescence-guided techniques, particularly 5-ALA, have been widely adopted to enhance intraoperative tumor

visualization and facilitate more extensive resection [8,9]. While prior studies have consistently demonstrated improved extent of resection with fluorescence guidance, the potential trade-off between surgical aggressiveness and neurological function has remained incompletely defined [8,41,44]. Our findings directly address this concern by demonstrating that enhanced visualization does not translate into increased postoperative neurological deficits, supporting the safety of fluorescence-guided approaches.

From a clinical perspective, these findings are highly relevant to surgical decision-making. The absence of a clear increase in neurological morbidity suggests that fluorescence guidance may enable surgeons to pursue more complete tumor resection with greater confidence, without substantially increasing the risk of functional impairment [8,9,16,21,41]. This is particularly relevant in modern neuro-oncology, where balancing maximal tumor control with preservation of quality of life remains paramount. By demonstrating comparable neurological outcomes, our findings support fluorescence-guided surgery as an adjunct that facilitates greater extent of resection while remaining consistent with the principle of maximal safe resection [8,18,41,42].

The consistency of findings across analyses further strengthens the validity of our results. Despite variation in study design, patient populations, and definitions of neurological outcomes, effect estimates were consistently centered around the null, with minimal to moderate heterogeneity, supporting the robustness of the observed associations. Larger studies contributed the greatest statistical weight and demonstrated similar neutral effect estimates, suggesting that the overall findings are unlikely to be driven by small-study effects or outlier results. Importantly, subgroup analysis showed no significant differences between randomized and observational studies, indicating that real-world data are concordant with controlled trial evidence.

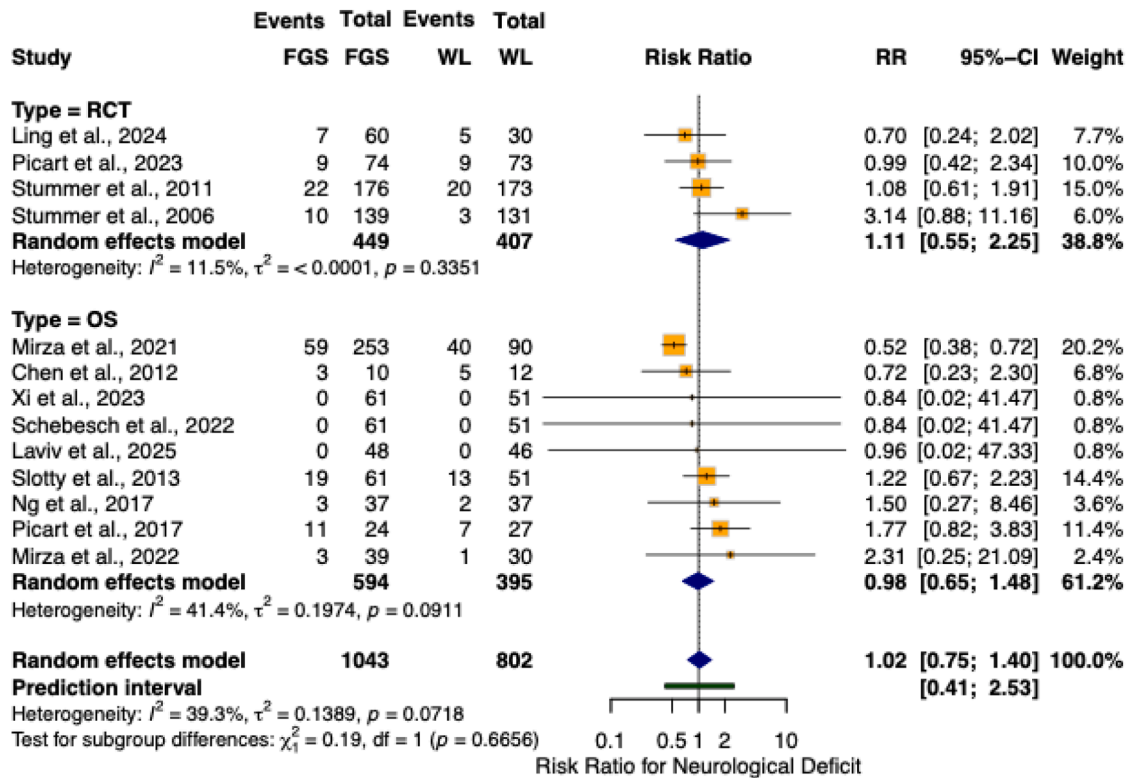
Notably, point estimates in some analyses suggested a non-statistically significant trend toward higher rates of neurological deficits in the fluorescence-guided group. However, these estimates were imprecise, with wide confidence intervals crossing the null, and likely reflect differences in case selection and surgical complexity. In clinical practice, fluorescence guidance may be preferentially used in more challenging tumors or eloquent regions, introducing potential confounding by indication. Therefore, the observed trend should not be interpreted as evidence of harm but rather as a reflection of underlying clinical heterogeneity.

4.1. Limitations

Several limitations should be considered. First, a substantial proportion of included studies were observational, introducing potential bias due to confounding and selection effects. Second, the definitions of neurological deficits were not standardized across studies, with variability in reporting of transient versus permanent deficits and differences in functional outcome measures. Third, detailed data on tumor location, eloquence, and use of adjuncts such as intraoperative mapping were inconsistently reported, limiting the ability to perform more granular subgroup analyses. Finally, the certainty of evidence was graded as low due to risk of bias and imprecision, reflecting the limitations inherent to the available literature.

Despite these limitations, this study provides a comprehensive synthesis of neurological outcomes following fluorescence-guided glioma surgery. By prioritizing functional and neurological endpoints, our analysis addresses a critical gap in the literature, which has predominantly focused on oncologic outcomes. Importantly, while traditional measures such as neurological deficits and performance status remain central, there is increasing recognition of the role of patient-reported outcomes (PROs) in capturing the broader impact of surgical interventions, particularly in neuro-oncology, where quality of life is a key determinant of treatment value [45,46]. Patient-reported outcomes capture domains such as quality of life, functional independence, and symptom burden that are not fully reflected by clinician-reported scales

A.



B.

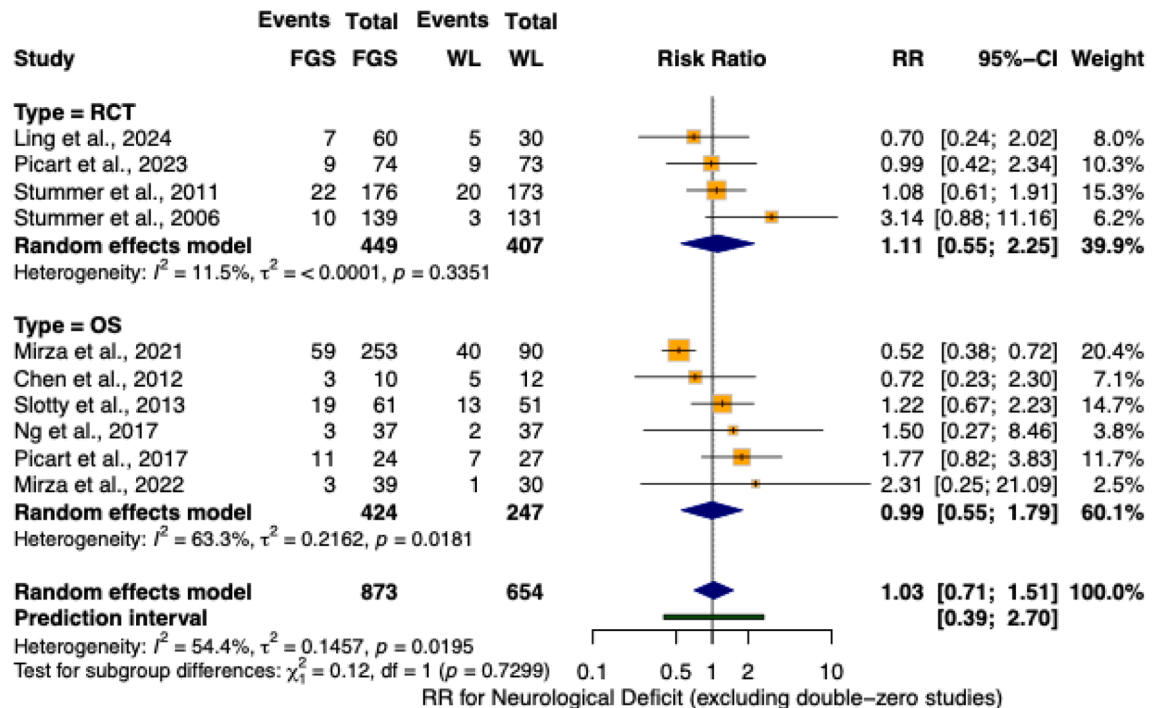


Fig. 3. Subgroup Analysis of Postoperative Neurological Deficits by Study Design. Forest plot showing subgroup analyses of postoperative neurological deficits comparing fluorescence-guided surgery (FGS) and white-light (WL) surgery, stratified by study design (randomized controlled trials [RCTs] and observational studies [OS]). Both A) primary and B) sensitivity analyses (excluding studies with zero events in both groups) are presented. Pooled risk ratios (RRs) with 95% confidence intervals (CIs) were calculated using a random-effects model. The test for subgroup differences was performed to assess interaction between study design and treatment effect. Study weights are indicated by square size, and diamonds represent pooled estimates for each subgroup and overall effect.

alone. Integration of patient-reported outcomes into future studies will be essential to better characterize the patient-centered benefits and potential trade-offs of fluorescence-guided surgery [45–47]. The concordance of findings across multiple analyses in our study supports the conclusion that fluorescence-guided surgery offers improved intraoperative visualization without increasing neurological morbidity, while highlighting the need for more comprehensive outcome assessment in future research.

5. Conclusion

Fluorescence-guided surgery enables improved intraoperative visualization without increasing neurological morbidity, supporting its role in achieving maximal safe resection in high-grade gliomas. These results provide clinically relevant reassurance for its widespread adoption and underscore the need for future studies integrating patient-reported outcomes to better capture the full impact of surgical intervention.

CRedit authorship contribution statement

Victor M. Lu: Conceptualization, Supervision, Validation, Writing – review & editing. **Ricardo J. Komotar:** Conceptualization, Project administration, Supervision, Validation, Writing – review & editing. **Vidya Govindarajan:** Data curation, Methodology, Validation, Writing – review & editing. **Adam S. Levy:** Data curation, Investigation, Methodology, Validation, Writing – review & editing. **Alejandro N. Santos:** Data curation, Methodology, Supervision, Writing – review & editing. **Marylourdes Andreu:** Data curation, Investigation, Methodology, Validation, Writing – review & editing. **Sai Sanikommu:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Ethics approval

This study does not require ethics approval.

Consent to participate

NA

Consent to publish

NA

Clinical trial number

NA

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Declaration of Competing Interest

The authors have no relevant financial or non-financial interests to disclose.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.clineuro.2026.109436](https://doi.org/10.1016/j.clineuro.2026.109436).

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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